

Diabetic Prediction Project

Course: Introduction to Data Analytics (ECE3807)

Team: [Your names]

Date: [Date]

1. Introduction

Diabetes mellitus is a chronic condition affecting millions worldwide. Early detection is critical for initiating care and reducing complication risk. This project aims to develop a predictive model to identify individuals at high risk of diabetes using clinical features from the Pima Indians Diabetes Dataset.

2. Objectives

- Build a reliable binary classifier to predict diabetes outcome.
- Demonstrate the end-to-end data analytics pipeline: collection, cleaning, feature engineering, model selection, evaluation, and inference.
- Provide documentation and artifacts for reproducibility.

3. Dataset

Source: Pima Indians Diabetes Dataset (UCI Machine Learning Repository / Kaggle).

Description: 768 observations, 8 clinical features, and one binary outcome variable (**Outcome**) indicating presence (1) or absence (0) of diabetes. Features include pregnancies, glucose levels, blood pressure, BMI, age, insulin, skin thickness, and diabetes pedigree function.

4. Methodology

4.1 Data collection

The dataset was placed in `data/diabetes.csv`. `data_fetch.py` can download the file programmatically from the UCI repository if internet access is available.

4.2 Preprocessing and cleaning

- Replaced zeros in clinically impossible fields (**Glucose**, **BloodPressure**, **SkinThickness**, **Insulin**, **BMI**) with NaN, then imputed using the **median**.
- Splitting into train/test using stratified sampling to preserve class balance.
- Features were standardized using **StandardScaler**.

4.3 Feature engineering

- An interaction term **BMI_Age** was created: **BMI * Age** to capture the combined effect of weight and age.

- No categorical encoding was necessary — all features are numeric.

4.4 Modeling

We trained: - Logistic Regression (baseline) - Random Forest Classifier

Model selection used 5-fold stratified cross-validation with AUC as the scoring metric.

4.5 Evaluation

Primary metric: ROC AUC. Secondary: precision, recall, F1-score, confusion matrix. Visualizations include ROC curve and confusion matrix heatmap.

5. Implementation (code)

Scripts are modular: - `data_fetch.py` — dataset download / sample creation - `data_preprocessing.py` — cleaning, imputation, scaling, splitting - `train_model.py` — training & model selection - `evaluate.py` — compute metrics & plots - `inference.py` — single-sample predictions

(Full code is provided in the project repository.)

6. Results

Insert actual numbers from model run here. Example (illustrative):

- Logistic Regression: CV mean AUC = 0.79 (std 0.03) - Random Forest: CV mean AUC = 0.81 (std 0.02) - Chosen model: Random Forest. Test AUC: 0.80. Precision: 0.75. Recall: 0.70.

Confusion matrix interpretation: [explain false positives vs false negatives and their clinical implications — e.g., a false negative is more serious clinically.]

7. Discussion

- The Random Forest model produced higher AUC than logistic regression (example).
- Limitations:
 - Pima dataset is not large and may not generalize across populations.
 - Many measurements are cross-sectional; longitudinal prediction would be better.
 - Imputation with median is a reasonable baseline but may mask distributional patterns.

8. Future Improvements

- Use additional datasets and external validation.

- Try ensemble of more models (XGBoost, LightGBM), hyperparameter tuning with GridSearch/Optuna.
- Model calibration for probabilistic outputs.
- Deploy as a web dashboard for clinicians and collect real-time data (bonus marks).

9. Conclusion

The project successfully demonstrates an end-to-end machine learning pipeline for diabetes prediction. With further validation and richer datasets, this pipeline can be extended toward a deployable clinical decision support tool.

10. Team contributions

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References

- UCI Machine Learning Repository: Pima Indians Diabetes Database.
- Scikit-learn documentation.